SGLang self benchmark design

AISim integration proposal | 10 November 2026

Preserve the vLLM self-benchmark workflow and AISim profile contract through an AISim-owned controller and a narrow SGLang execution adapter. SGLang continues to own model execution, allocation and native cache operations. Benchmark workers may exit after collection.

The design targets no changes to normal SGLang serving paths. Function-level reuse is supported by the inspected source, but the general adapter and full feature parity still require implementation and GPU qualification.

Goals and boundaries

Preserve	Explicit boundary
Pure prefill and pure decode; combined mode runs the pure phases in sequence.	Same workload geometry and state provenance, with actual shape checks.
Planning, attempts, smoke, resume, retries and profile publication.	Extend backend contracts without weakening vLLM validation.
Warmup, graph-aware sampling, raw evidence and supported state modes.	Full parity is the target; a TP-only first stage is incomplete.
Disposable benchmark worker.	Exact scheduler timing and same-worker serving need separate integration.

Existing vLLM lifecycle

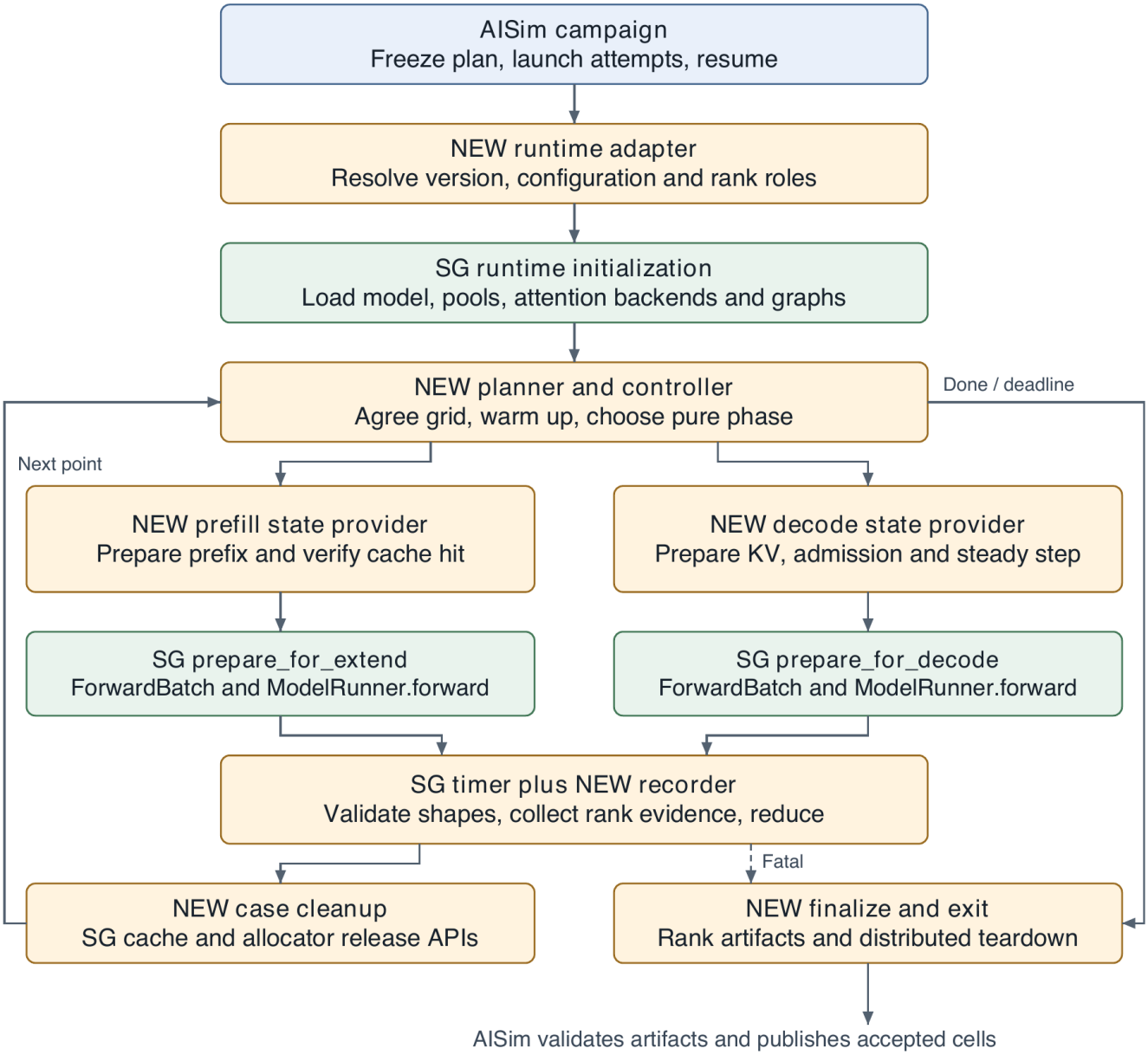
AISim normally launches separate prefill and decode cells. Dynamo can also run both sweeps in one worker with benchmark mode agg. In either case the phase sequence follows the same contract. [1-6]

Stage	Current behavior
Plan and start	Freeze runtime identity, create an attempt, load model and resolve actual capacity.
Grid and warmup	Admit explicit points or build a capacity-aware grid; discard warmup measurements.
Pure phases	Prepare cache state outside timing; measure prefill or steady decode at verified coordinates.
Finalize and collect	Drain work, write rank evidence, merge results and restore native worker state. AISim then stops the engine.
Validate and publish	Check coverage, provenance and timing; publish Parquet and its metadata commit record.

Version choice: use v0.5.22 as a provisional customer baseline and qualify v0.5.23 with the same adapter. Confirm the actual customer image during qualification. Their inspected one_batch.py files are identical, but the internal interfaces have no stable public API guarantee. [11]

Proposed execution flow

Ownership is explicit in every block. Blue identifies reusable AISim machinery, green existing SGLang functions, and amber new controller or adapter code. Mixed blocks name both responsibilities.

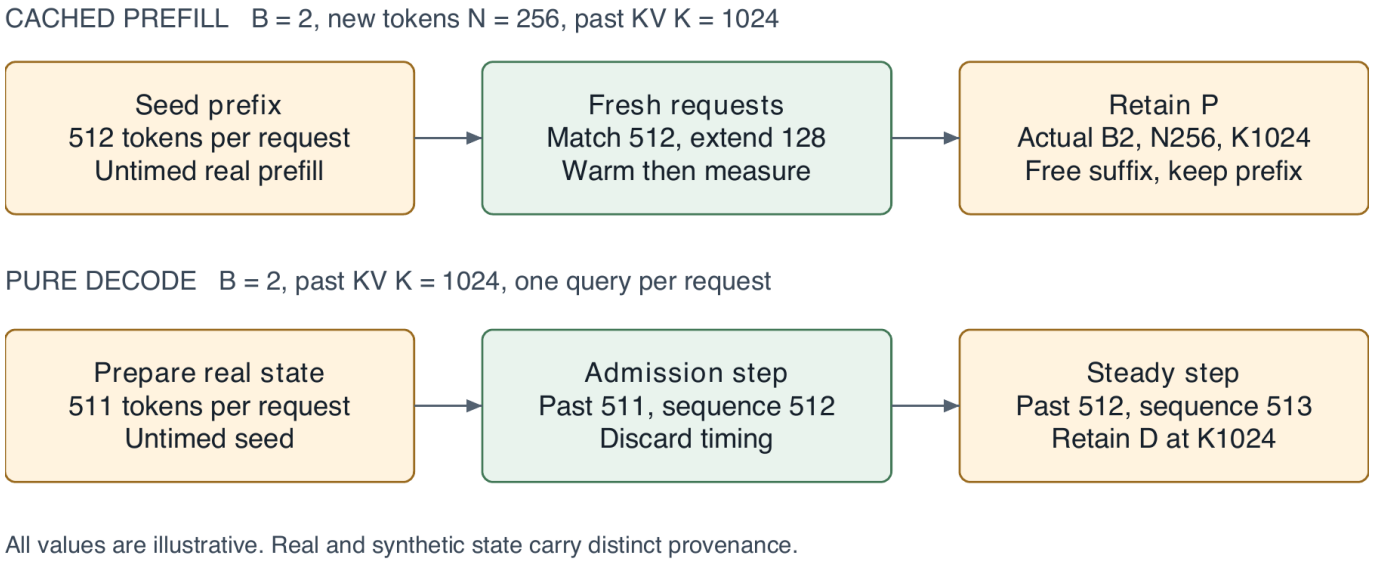


The loop releases each case before the next point. Fatal failures finalize an invalid attempt; generated-point skips can continue but invalidate complete coverage. Process groups are destroyed only at worker exit.

The final AISim stage checks completion and provenance before applying campaign publication policy. A fresh serving worker can start afterward. Reusing the benchmark worker for serving is outside the accepted disposable-worker baseline.

State preparation and worked examples

B counts requests; N sums new prefill tokens; K sums past KV tokens. These are totals within one attention-DP rank, not across all ranks. Decode has N = 0 but processes one query per request. The examples assume full attention, page size 16 and sufficient capacity. [2-4, 7, 12]



Cached prefill

Seed the prefix through an untimed real prefill, checkpoint it, then create fresh requests and match the native cache. Require exactly 512 cached tokens for each 640-token request. Execute the remaining 128 tokens per request and release measured suffixes without inserting them into the cache. This keeps later warm or measured shots from accidentally hitting the entire prompt.

Use the native cache builder and registry so the selected implementation matches the model and runtime configuration. one_batch.extend uses a dummy cache; its helper alone does not provide insertion, lookup, locking or eviction parity. [11, 13-14]

Steady decode

vLLM normally discards an admission step and selects the following steady step. The example uses a real 511-token seed to reproduce that geometry in SGLang. The second decode reads 512 past tokens per request, so K = 1024; its attention sequence length is 513, including the query. That crosses from 32 to 33 pages per request. State provenance remains distinct from synthetic admission. [4, 7, 12]

State and ownership invariants

- Maintain output token IDs, finish state and cache locks. SGLang allocation already advances KV counters; do not advance them again after forward.
- Never checkpoint a failed forward. Release requests through native helpers, drain GPU work and create fresh request objects. Resetting allocator free lists alone is insufficient.
- Real prefix, synthetic KV, exact recurrent state, randomized recurrent state and approximate live-state modes need separate providers and eligibility checks. Recurrent snapshots are not interchangeable with immutable attention KV.

Component responsibilities and interfaces

Keep orchestration in AISim and isolate SGLang internals behind an adapter. The names below define proposed module boundaries; they are not APIs already implemented in the repositories.

Component	Reuse	New construction
Campaign backend	AISim plans, deployment, attempts, resume and publication.	SGLang admission, launch arguments, probe and producer contract.
Runtime adapter	Configuration resolution, bootstrap and one_batch.load_model.	Correct initialization order and rank roles; preserve customer graph configuration.
Controller and planner	vLLM phase protocol and point schema; native pool and graph limits.	Ordered phases, capacity admission, uneven per-request shapes, warmup and deadlines.
State providers	Native cache builder, match/checkpoint/lock/release and pools.	Prefix and decode preparation, model-specific state modes and lifetime ownership.
Execution adapter	ScheduleBatch, prepare_for_extend/decode, ForwardBatch, ModelRunner.forward.	Assemble calls, consume output, update request state and retain dispatch evidence.
Recorder and coordinator	DeviceTimer, distributed groups and MLP synchronization.	Sample identity, timer draining, rank agreement, reductions and failure propagation.
Artifact adapter	AISim validation, Parquet writer and consumers.	SGLang runtime metadata and explicit state/timing policies with reader tests.

Small adapter surface

The controller calls initialize(config), observe_limits(), prepare(case), execute_step(state), collect_sample(), release_case(state) and close(). These conceptual methods keep backend state transitions out of campaign and publication code.

Records crossing the boundary

Record	Required contents
CaseSpec	Phase; B, N, K within one attention-DP rank; per-request lengths; input seed; state, warmup and reduction policies.
RuntimeIdentity	Image/revision, model, precision, topology, resolved backends and graph configuration, observed capacity.
RawSample	Attempt/run/point/rank/step IDs, actual shape, timing kind and duration, state regime, graph evidence.
CaseResult	Retained sample or reduction, complete raw evidence, validity, failure reason and coverage status.

Retain plan hash, grid digest and attempt identity through serialization. Use deterministic inputs shared within TP groups and the intended distinct content across attention-DP ranks. [2, 5, 8-10]

Measurement and reliability contracts

Timing and graph evidence

Metric	Boundary and consequence
vLLM scheduler wall	Prefill schedule-to-output or steady decode output-to-output. It includes scheduler behavior and asynchronous execution effects.
SGLang operation wall	Synchronized duration around the standalone operation. It is a different interval from scheduler wall time.
SGLang native GPU span	Existing DeviceTimer interval. Eager and graph boundaries differ; graph timing can include batch loading. Sampling is outside the native forward timer.

Attach the existing timer hook, bind intervals to a sample ID, synchronize and drain before moving to another sample. Preserve raw timings and declare the selected policy. A new standalone policy needs writer and consumer support; neither the generic vLLM label nor the existing model-specific SGLang median label is a generic substitute. [5, 16-17]

Retain ModelRunnerOutput.can_run_graph. Record expected graph buckets separately from observed dispatch; a general observed prefill bucket accessor was not found. Exact bucket evidence may justify an optional upstream metadata extension. Stock one_batch.main changes graph maximum batch size and must not silently control customer graph policy. [11, 15]

Warmup and reduction

Record actual warmup execution, not just requested counts. Keep admission, warm shots and measured samples separate. vLLM defaults to one independent repetition; some state policies reduce several adjacent steady steps with growing context. Preserve that declared behavior. Fixed-context independent repetitions are a separate policy. [3-5, 18]

Rank coordination and failure handling

- Agree on capacity, point order, grid digest and state eligibility before coupled execution. Distinguish TP/attention-DP/EP participants from independent replicas. The vLLM reference takes the maximum attention-DP rank duration; each SGLang timing policy must specify its own valid group reduction.
- An invalid explicit point fails. A generated-point skip records missing coverage. Neither can silently become a complete accepted cell.
- Stop soft timeouts at coordinated boundaries. Bound finalization, preserve raw evidence and logs, and clean up on success or failure. Cleanup failure fails the attempt.
- AISim rejects incomplete native cells for complete-cell publication. Explicit partial-campaign publication can include complete passed cells while listing missing cells. Resume remains bound to the frozen plan and attempt provenance. [8-10]

Exact scheduler-wall parity requires a scheduler execution driver with controlled admission and output observation. A timer change in a standalone runner cannot provide it.

Implementation stages and acceptance

Implement the contract incrementally while retaining the complete feature inventory. Each stage has an observable acceptance condition; completion of the first stage does not imply full vLLM parity.

Stage	Deliverable	Acceptance evidence
1 Contracts	Backend-neutral records, explicit SGLang policy and isolated runtime adapter.	CPU contract tests cover geometry, plan/attempt identity, completion and rejected unsupported modes.
2 Reference execution	TP ordinary autoregressive models; real-state cached prefill and steady decode.	GPU traces verify actual B/N/K, admission exclusion, cache hits, graph/eager paths and leak-free repetition.
3 Campaign integration	Probe, raw artifacts, writer/consumer support, smoke, retries and resume.	End-to-end collection publishes a valid table; injected failures preserve evidence and never publish invalid cells.
4 Complete parity	Uneven shapes, qualified synthetic/recurrent modes and supported distributed topologies.	Feature matrix is satisfied per model/layout; group failures, capacity boundaries and state regimes are tested.
5 Version qualification	Confirm the customer image; provisionally target v0.5.22 and qualify v0.5.23.	Same contract suite passes; controlled serving comparisons quantify timing differences for each accepted policy.

Release acceptance checklist

- Preserve point totals and per-request partitions; reject or explicitly classify unrepresentable cache geometry instead of silently rounding it.
- Verify cached prefill, decode off-by-one arithmetic, page boundaries, graph capture boundaries and eager tails.
- Confirm no measured sample contains setup, admission or warmup unless its declared timing policy intentionally includes it.
- Require complete rank evidence, matching identities, truthful state provenance and successful cleanup before accepting a cell.
- Qualify each supported cache layout and topology; compare representative points with ordinary serving under matched runtime settings.

Decisions retained in the design

Disposable workers are allowed. Same-worker restoration is therefore outside the baseline; a fresh serving worker is an optional handoff. If exact scheduler-wall timing or same-worker reuse becomes mandatory, add a separate scheduler integration before claiming that parity.

Keep the current mode-specific vLLM restrictions, including decode/agg exclusions for speculation and pipeline parallelism. Multimodal, LoRA and other backend capabilities are not automatically covered by self-benchmark support.

Source map and review baseline

The design is grounded in source and regression-test inspection. It does not establish runtime compatibility or prediction accuracy. GPU qualification and controlled serving comparison are release requirements.

Repository	Inspected revision
Dynamo	33cfa1271c08f1660fc156ab2bc50647a268b87e
vLLM	v0.33.0 / 4ba787ec9959e566d3b587a796dff07673324da5
AISimulate	ba02b3071c1ed09e65c7d8682e959c292dedb677
SGLang	v0.5.22 and v0.5.23; implementation references below use v0.5.23.

[1] Dynamo opt-in scheduler and benchmark activation

[2] Point schema and per-request partitions

[3] Capacity-aware grid and phase controller

[4] Steady decode construction and context accounting

[5] Measurement protocol and raw evidence

[6] Worker finalization and restoration

[7] vLLM past context plus query sequence length

[8] AISim campaign attempts and execution

[9] Native artifact validation and complete coverage

[10] Profile row construction and publication

[11] SGLang model loading and benchmark helpers

[12] SGLang decode preparation and sequence advancement

[13] Model-specific native cache construction

[14] Cache release and ownership contract

[15] Forward result and graph dispatch

[16] Native timer interface

[17] AISim consumer timing and state policies

[18] vLLM second-step selection regression test

Primary compatibility risks are private initialization APIs, model-specific cache ownership, recurrent-state preparation, and timing boundaries. Keep those dependencies inside the adapter and require explicit version qualification.